{EPITECH}

<SINCE 1999/>

< Robotique TEK5 · Day 3 · Visio />

Perception: from pixels to 3D

< Calendar />

	Group A - October	Group B - Late November
Kickoff + Bootstrap	8 & 9 Oct	26 & 27 Nov
Visio #1 · Today	**22 Oct**	**10 Dec**
Intermediate review	~31 Oct	~19 Dec
Visio #2	5 Nov	7 Jan
Final review	21 or 28 Nov	23 or 30 Jan

Next deadline: intermediate review → Phase 1 working in sim, ideally on the real arm.

< Where you are />

1 driver

`/joint_command → arm`

2 brain / IK

`target → joint angles`

3 perception

`camera → (x, y, z)`

TODAY

PHASE 1

See the object.
Grasp it.
Drop it.

PHASE 2

Your application

< Today />

See in 3D

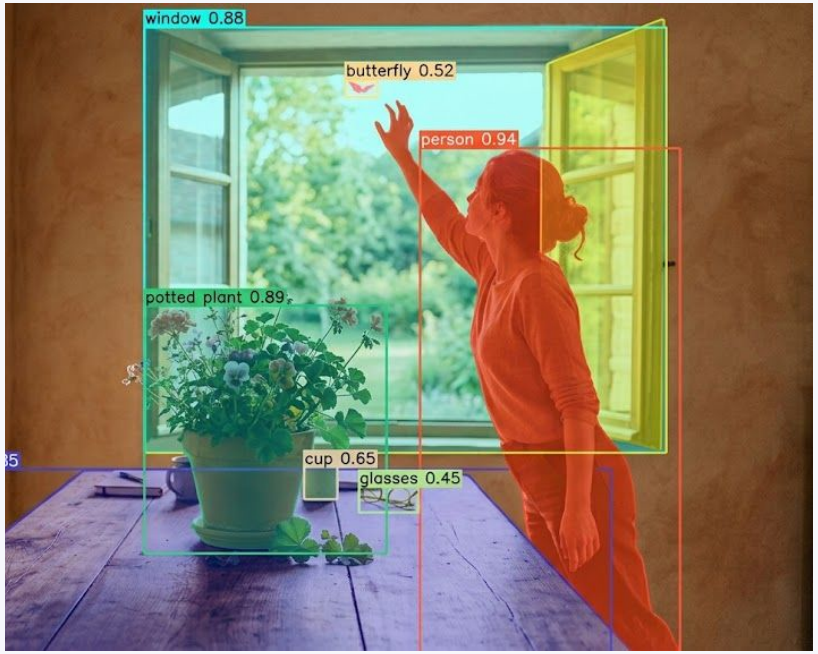
Phase 1





Guess depth

Find anything





Track humans

Ideas for phase 2.

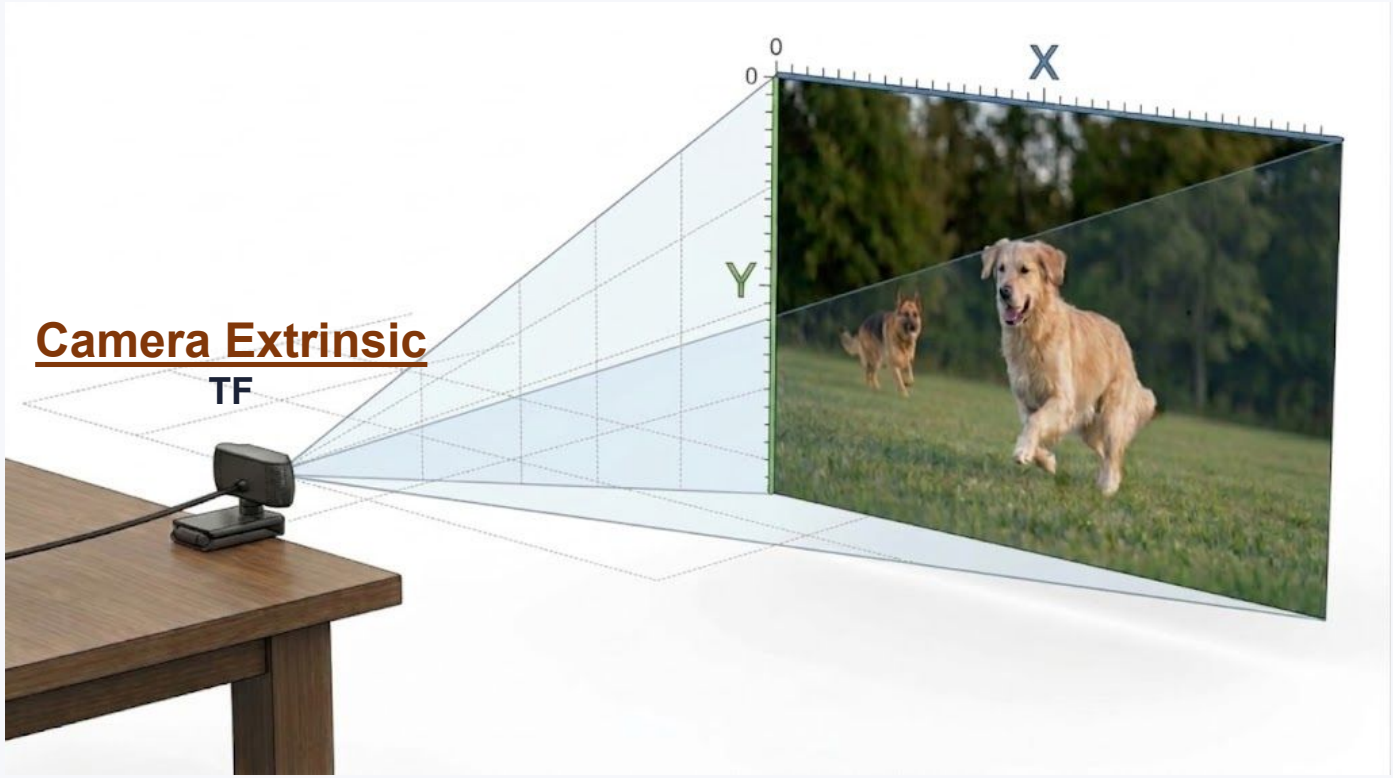
{EPITECH}

<SINCE 1999/>

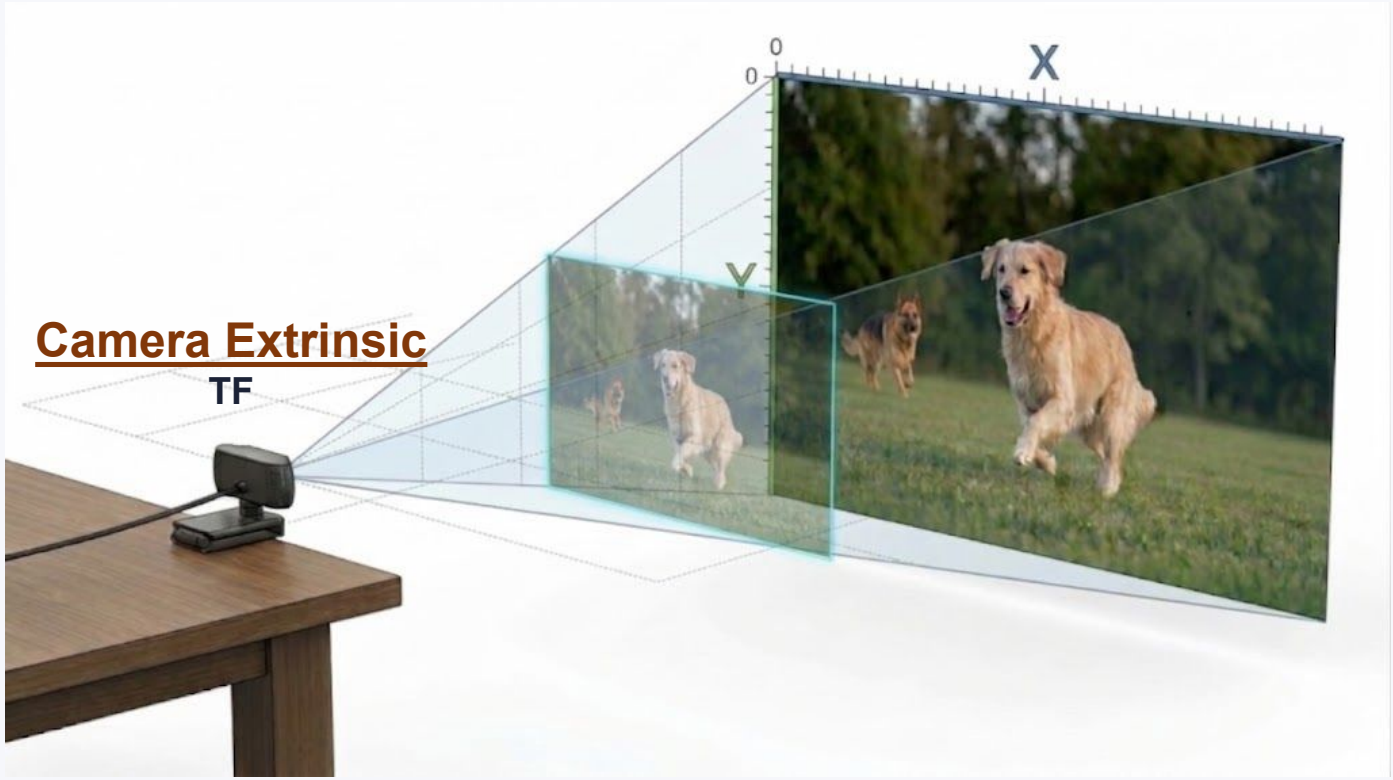
< Robotique TEK5 · Day 3 · Visio · Part 1 />

Geometry: from pixel to 3D

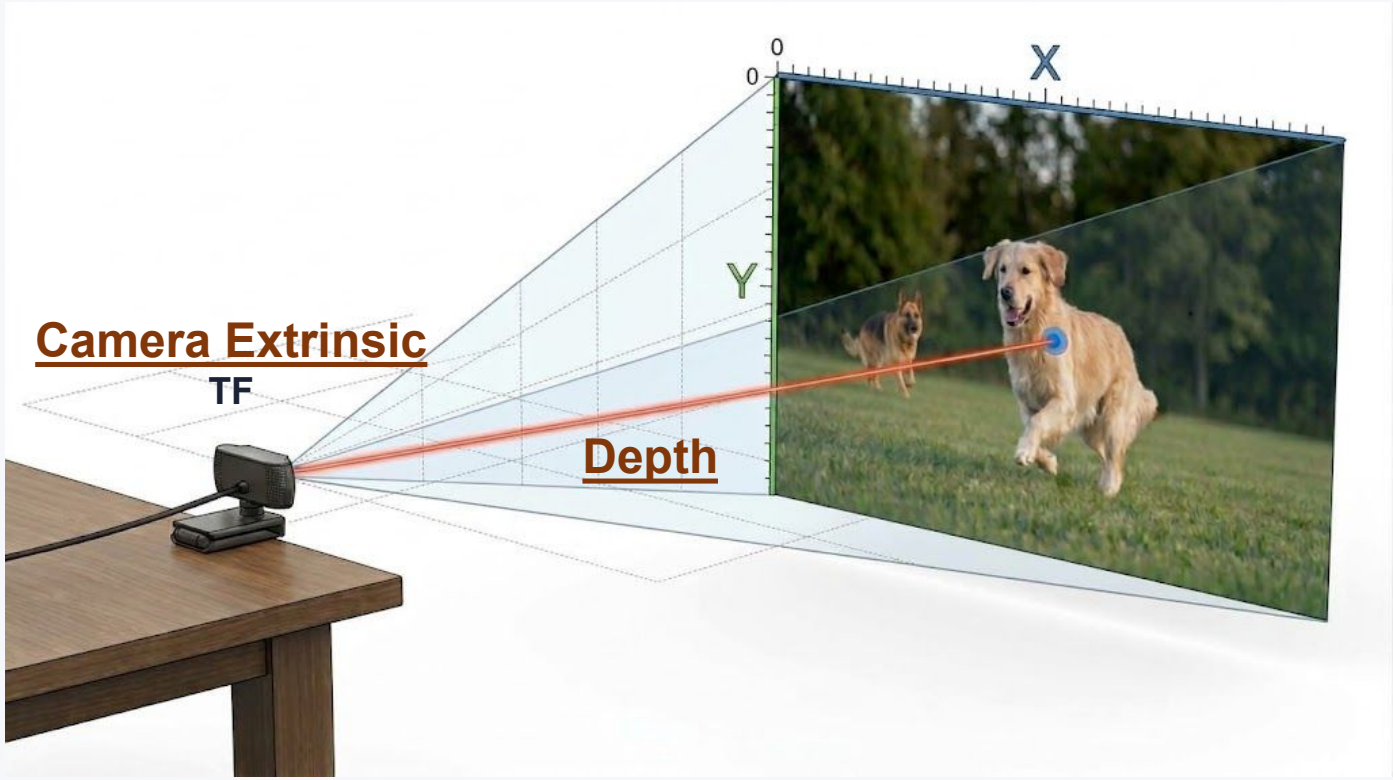
< What we need to place a pixel in 3D />

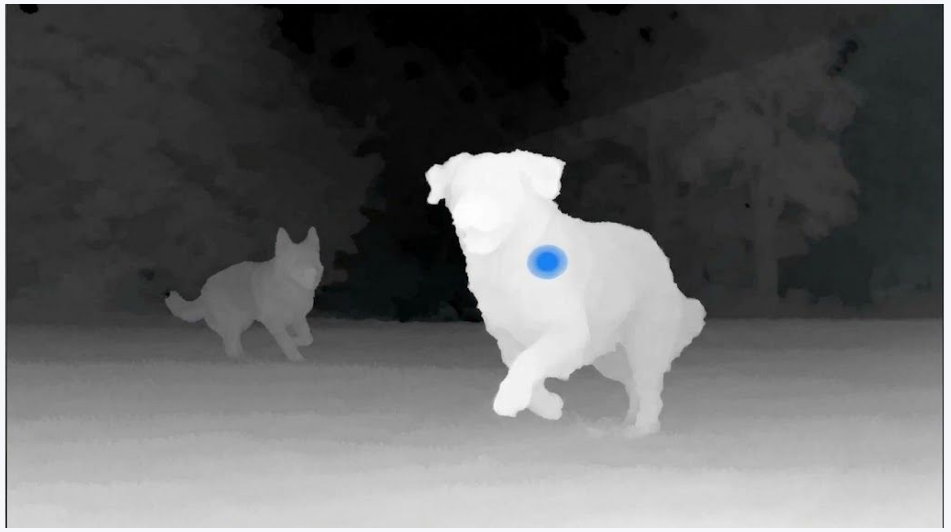


< What we need to place a pixel in 3D />



< What we need to place a pixel in 3D />



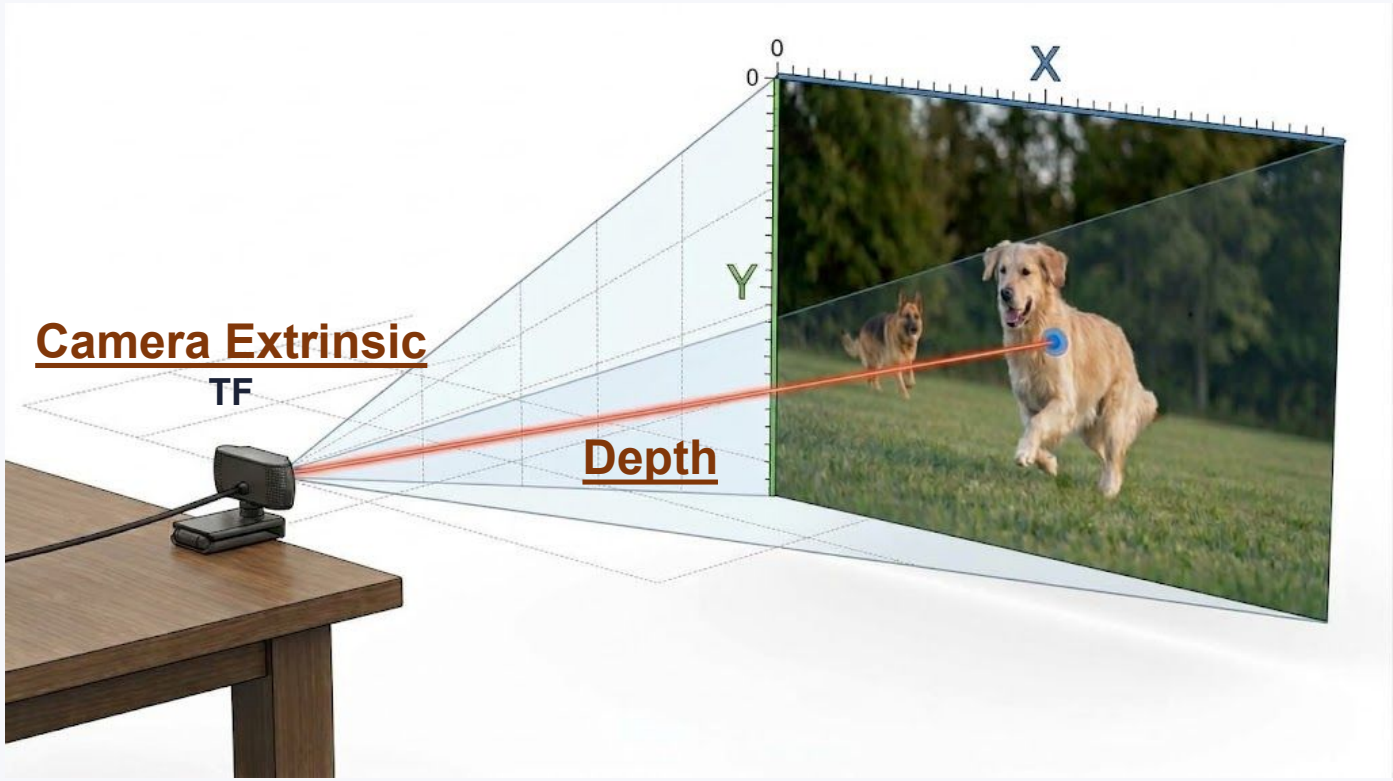


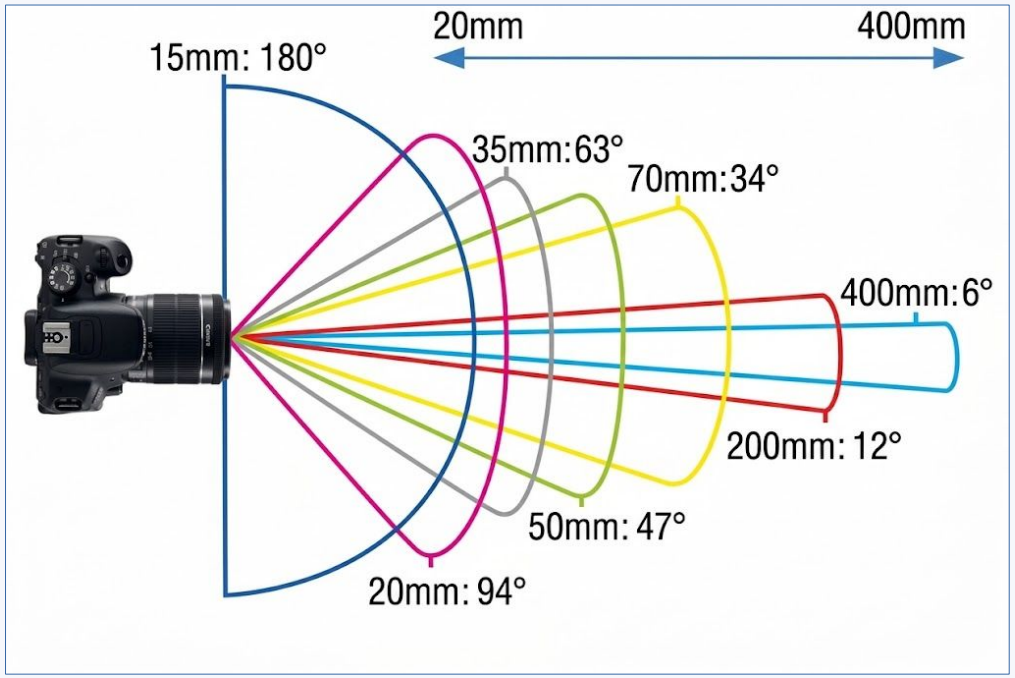
For now, imagine that you have this depth map

What's the depth for each image pixel ?

< What we need to place a pixel in 3D />







Camera Intrinsics

K (FOV & Focal)

D (distorsion lentille)

< What we need to place a pixel in 3D />

D

pixel
(u, v)

→ K

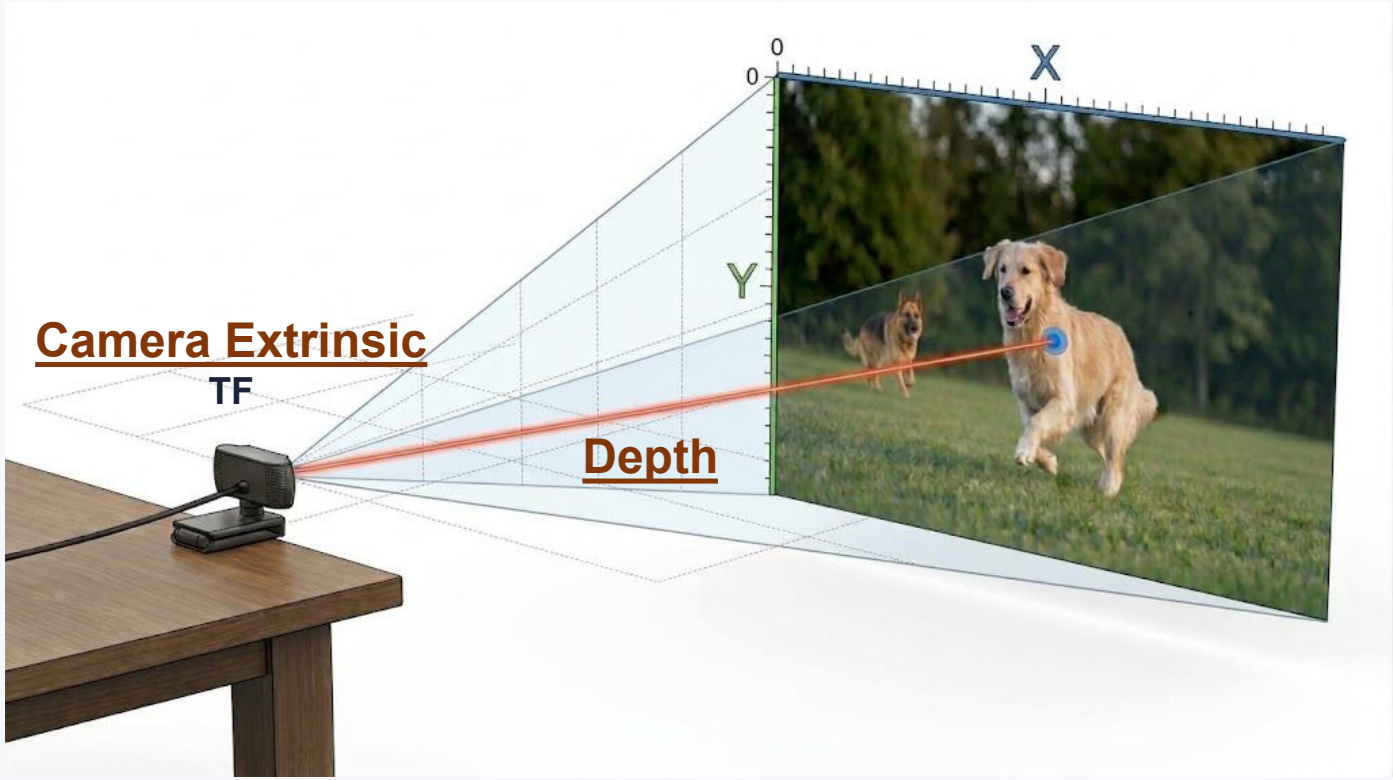
ray
$K^{-1} \cdot [u\ v\ 1]$

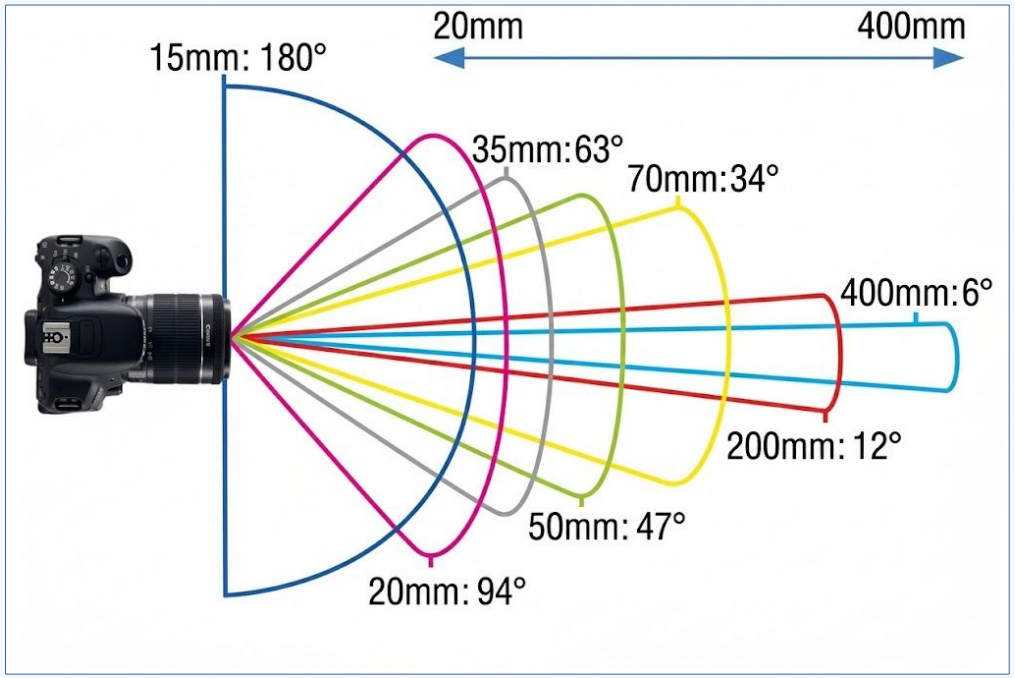
→ Depth

camera frame
(X, Y, Z)

→ TF

base frame
T_base_cam





Camera Intrinsics
K (FOV & Focal)
D (distorsion lentille)

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< Robotique TEK5 · Day 3 · Visio · Part 1 />

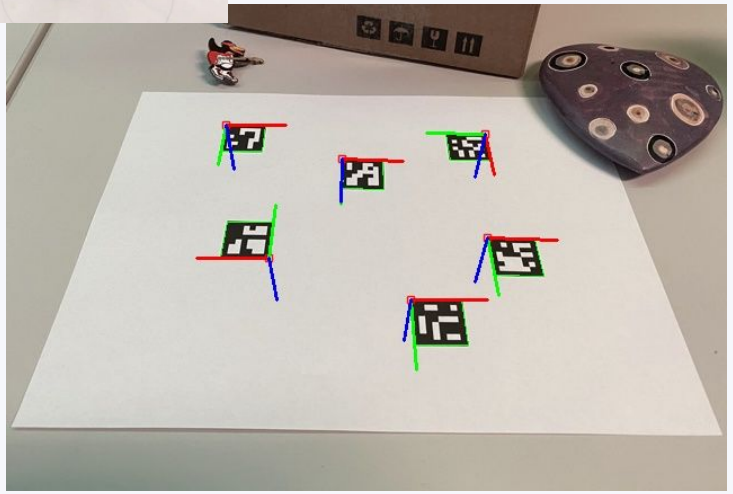
Calibration & camera position

< Camera in your project />

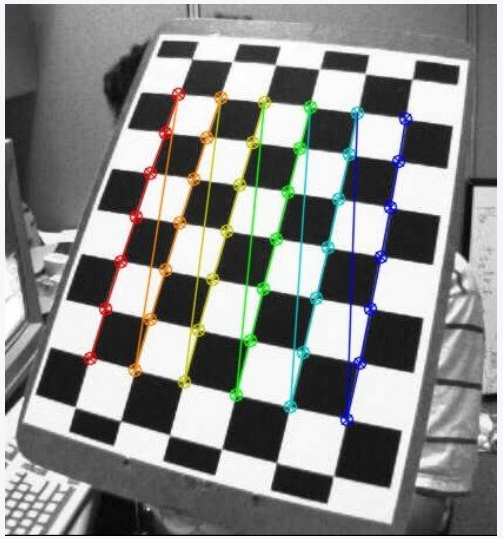


Use whatever camera you have at your disposal but...

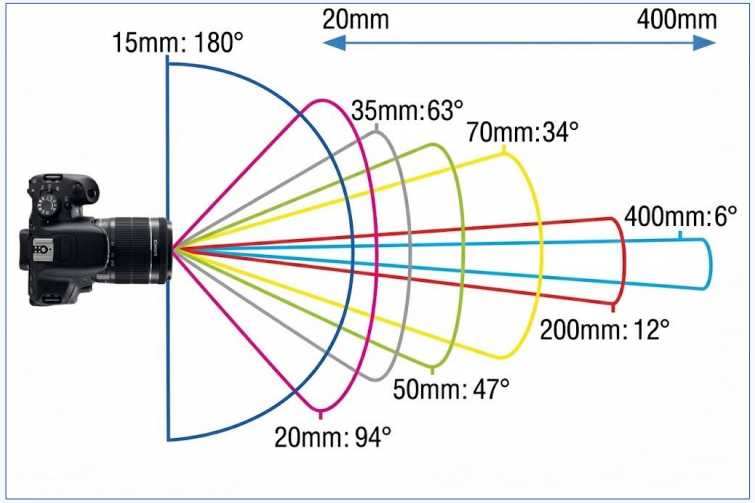
Reality : **K** + **D** unknown?



Estimate camera pose (TF)
with ArUco for ex…



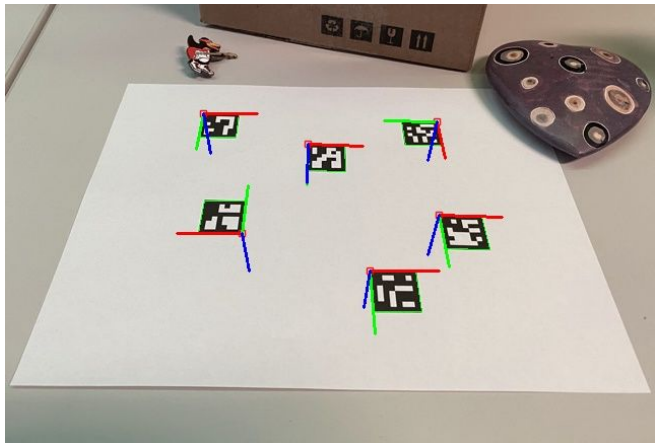
Calibration !
Watch some videos about it !



Autofocus OFF
or K and D will always change !

```
ros2 run camera_calibration cameracalibrator --size 8x6 --square 0.025
```

< Estimating robot and camera TFs />



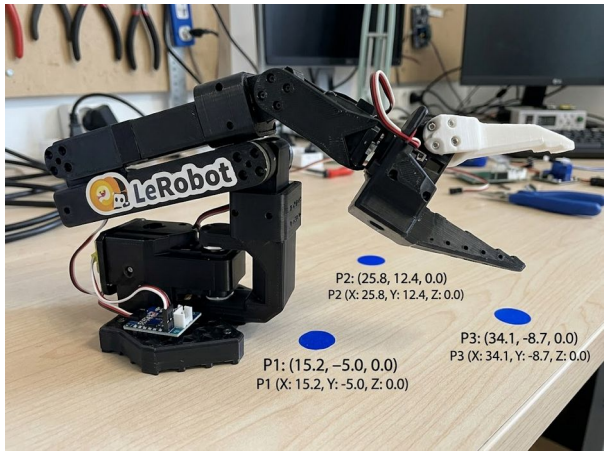
Marker on the table

Solve **PnP**



Hand-eye

AX = XB



Home-made

PnP (Perspective-n-point)
+
FK (Forward Kinematic)

→ **one static TF : *robot base_link* → *camera***

{EPITECH}

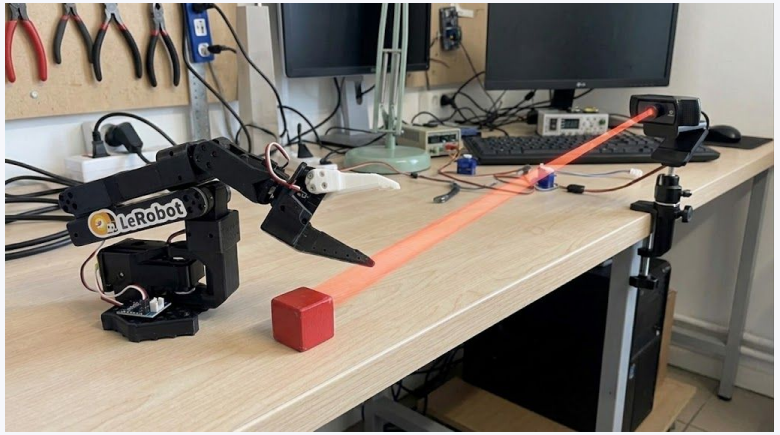
<SINCE 1999/>

< Robotique TEK5 · Day 3 · Visio · Part 2 />

Depth: sensors and AI

{ <TECH TOGETHER TOMORROW> }

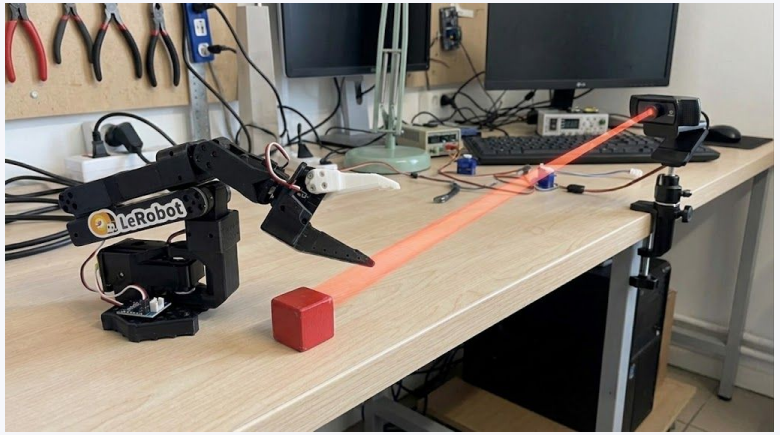
< RGB camera & geometry : how far? />



Known plane

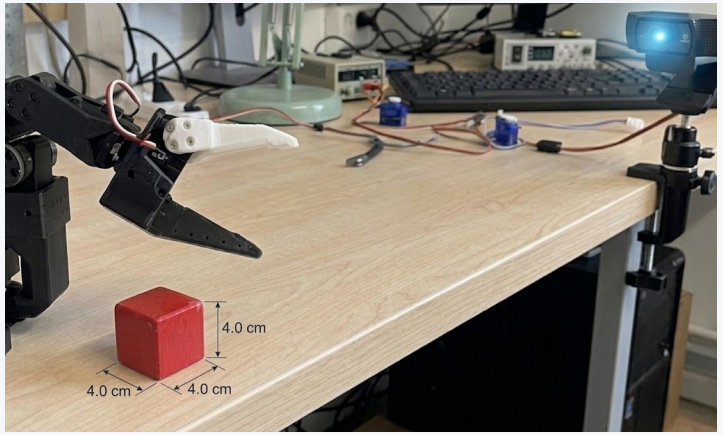
`ray ∩ table`

< RGB camera & geometry : how far? />



Known plane

`ray ∩ table`



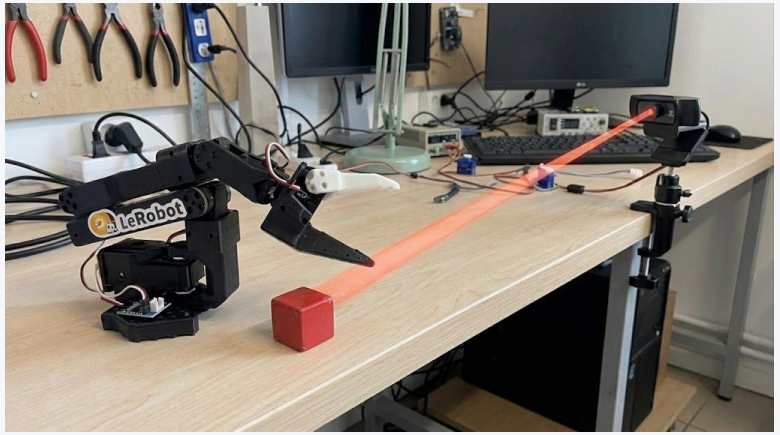
Known size

`then estimate depth using camera's parameters (simple geometry)`

Simplifications

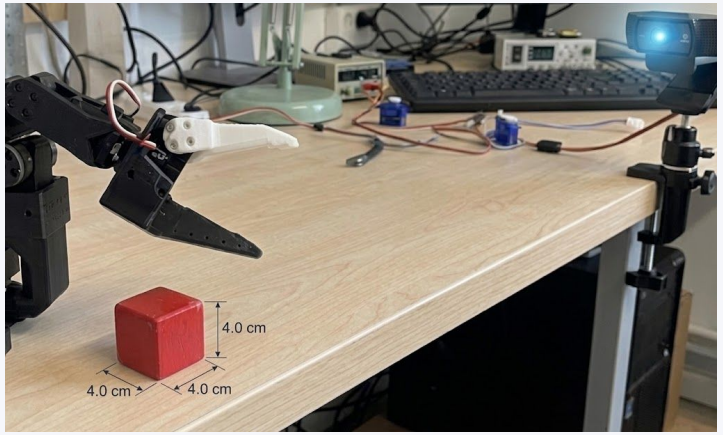
best for phase 1 ?

< RGB camera & geometry : how far? />



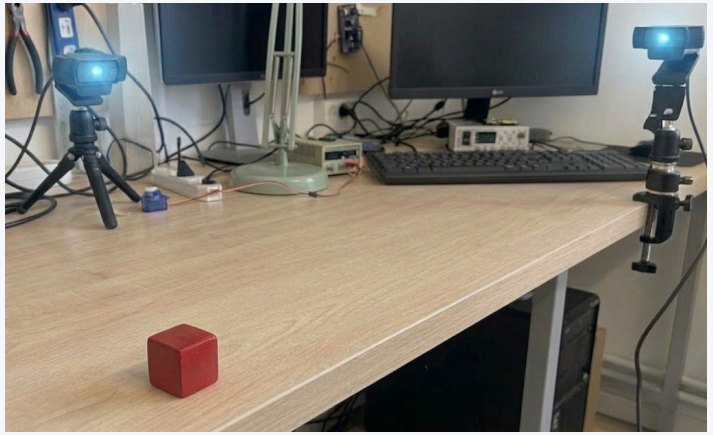
Known plane

`ray ∩ table`



Known size

`then estimate depth using camera's parameters (simple geometry)`



Several views

`Triangulation (~stereoscopy)`

Simplifications

best for phase 1 ?

< Depth cameras : RGB-D />



Passive stereo

ZED, OAK-D lite

~ 30-350 $



Active stereo

RealSense, Oak-D Pro

~ 350-600 $



Two images (stereoscopic)

Features matching to reconstruct depth → Expensive to compute :(

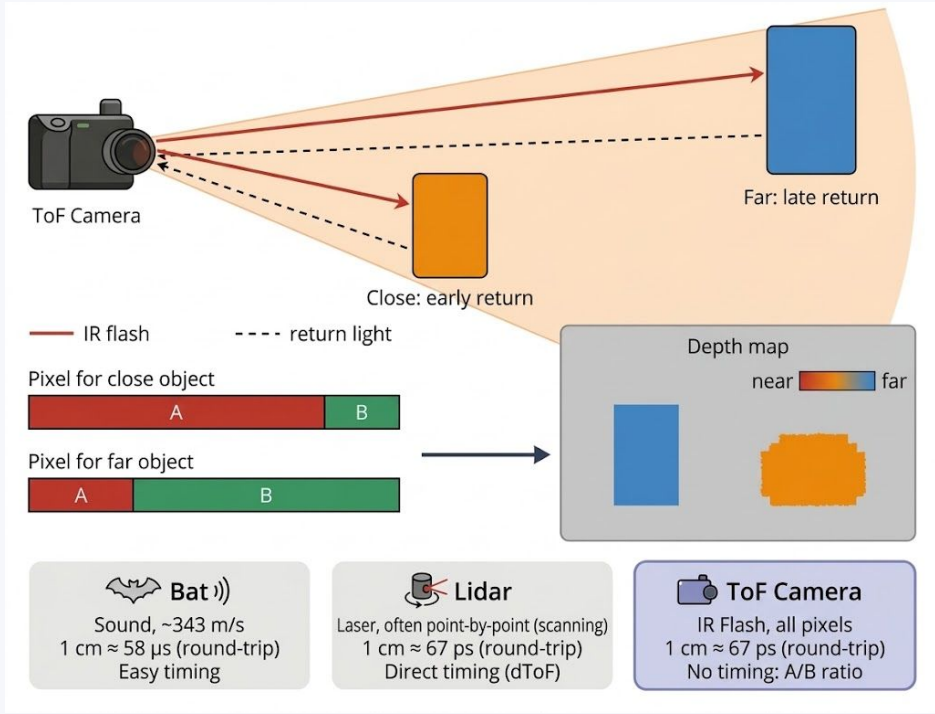
Close one eye: Stereo vision mimics our eyes. But do you lose depth?



Time of Flight

Orbbec Femto

~ 150-600 $



< Depth cameras : RGB-D />

https://docs.luxonis.com/overview/toplevel-features/depth/



Passive stereo

ZED, OAK-D lite

~ 30-350 $

++ price
++ outdoor
+ long range

−− needs texture 🧱
− needs light ☀️
− Accuracy/noise (error ∝ d²)
− Expensive compute 🖥️

~ Autonomous cars (tesla...)

https://www.realsenseai.com/get-started-depth-camera/



Active stereo

RealSense, Oak-D Pro

~ 350-600 $

−− pricier
+ outdoor (passive)
+ long range (passive)

+ blank surfaces ⬜
+ low light 🌙
− Accuracy/noise (error ∝ d²)
− Compute 🖥️

Robotics (unitree, robot arms...)

https://doc.orbbec.com/documentation/Orbbec%20Femto%20Bolt%20Documentation/Depth%20Camera%20(Femto%20Bolt)



Time of Flight

Orbbec Femto

~ 150-600 $

− pricier
−− inside mainly (sensitive to ambient light)
− limited range

++ blank surfaces ⬜
++ low light (even complete dark) 🌑
++ Accuracy/noise (~constant error)
++ Cheap compute 🖥️

Gesture tracking (Kinect v2) & Logistics (parcel sizing...)

< Close one eye />

Look far away.

How far is it?



Familiar size

Perspective

Occlusion

Texture

Shadows

Motion parallax

Experience = priors

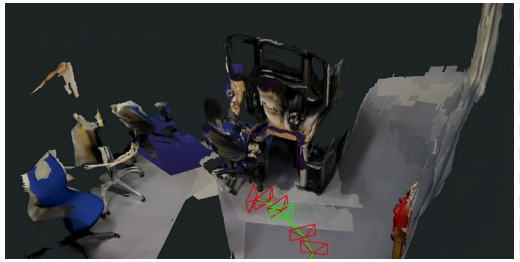
AI learns them

< Depth from one image using AI />



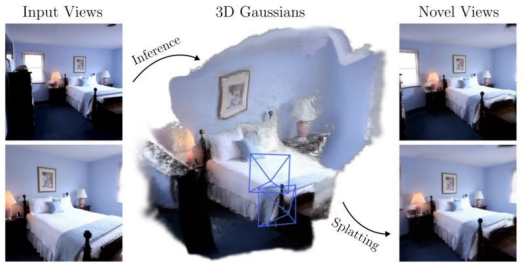
https://huggingface.co/spaces/depth-anything/Depth-Anything-V2

< Foundation AI: From visual intuition to real-world 3D />



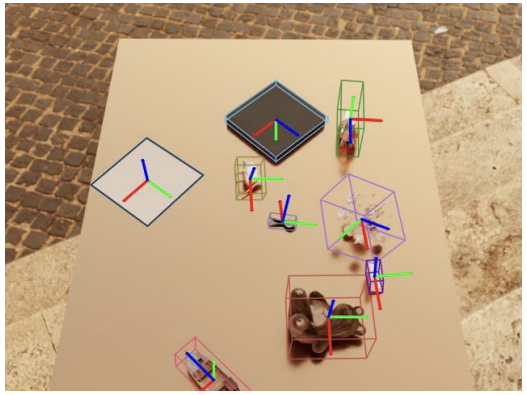
Dense 3D map built live from one camera
(Naver labs)

MASt3R-SLAM



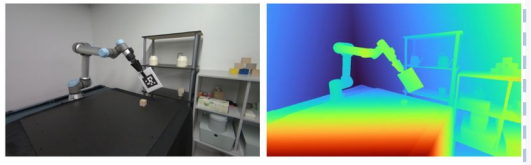
Gaussian splatting scene from a few photos
Video with postprocess

Feed-forward 3DGS



6D pose of an unseen object
(NVIDIA)

FoundationPose



Two cameras →
Clean stereo depth, zero-shot
(NVIDIA)

FoundationStereo

AI intuition unlocks real-world 3D…

… at a computational cost

< 3D foundation models in Robotics />

2024	2024	2025	2025	2026
DUSt3R	**MASt3R** Video Mast3er-slam	**VGGT**	**MapAnything**	**VGGT-Ω**

Naver labs Europe
Website



Images in → 3D points + cameras out. No calibration.

< Tesla: cameras only />



2021 **Vision only**

2022 **Occupancy Networks** – *IA perception + c++ for control*

2024 **FSD v12: end-to-end** – *full IA perception + control*
But black box !

2025 **+ 3D outputs, world simulator**
3D environment reconstruction

Waymo (their competitor), fusion : lidar + radar + cameras

< Tesla: cameras only />



2021 **Vision only**

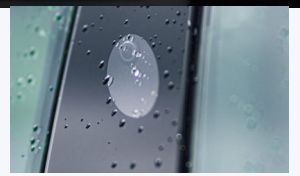
2022 **Occupancy Networks** – *IA perception + c++ for control*

2024 **FSD v12: end-to-end** – *full IA perception + control*
But black box !

2025 **+ 3D outputs, world simulator**
3D environment reconstruction



VS



− price (100-350 $)
++ outdoor
+ long range

−− needs texture 🧱
− needs light ☀️
− Accuracy/noise (error ∝ d²)
+ Low compute 🖥️

++ price (~15 $)
++ outdoor
++ long range

+ needs texture 🧱
− needs light ☀️
+ Accuracy/noise (error ∝ d²)
−− Massive compute 🖥️
→ they moved the cost from cameras to expensive chips for AI

Waymo (their competitor), fusion : lidar + radar + cameras

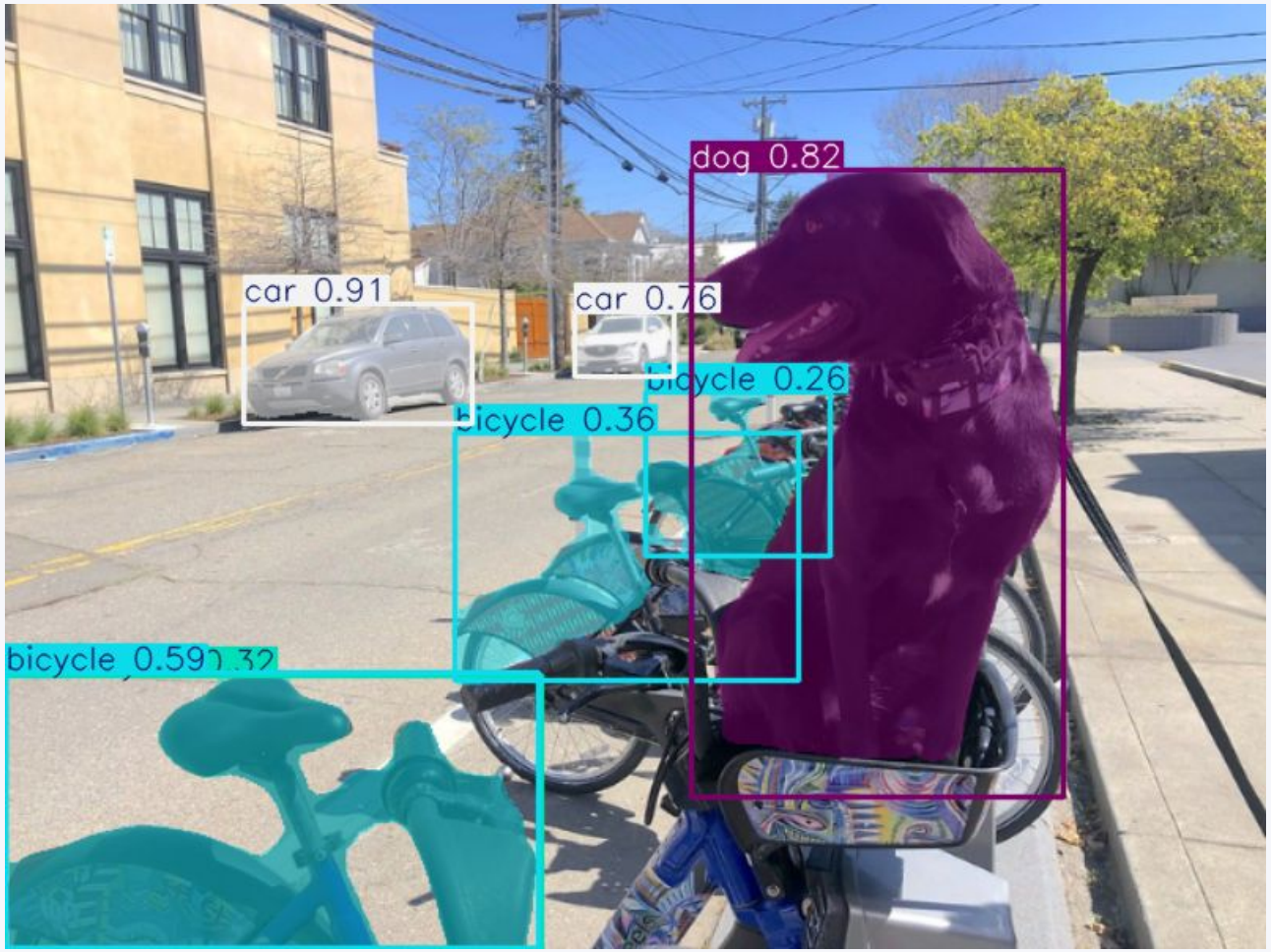
{EPITECH}

<SINCE 1999/>

< Robotique TEK5 · Day 3 · Visio · Part 3 />

Detection: YOLO, SAM3 and friends

< YOLO />



Real time (must be >60 fps)

Jetson nano → 2 to 5 fps :(
Jetson Orin nano → ~100-150 fps
Jetson AGX Orin → >300 fps

detect · segment · pose

YOLO26 (2026): NMS-free

License: AGPL-3.0

```
from ultralytics import YOLO
m = YOLO('yolo26n.pt')
boxes = m(frame)[0].boxes
```

Ultralytics website

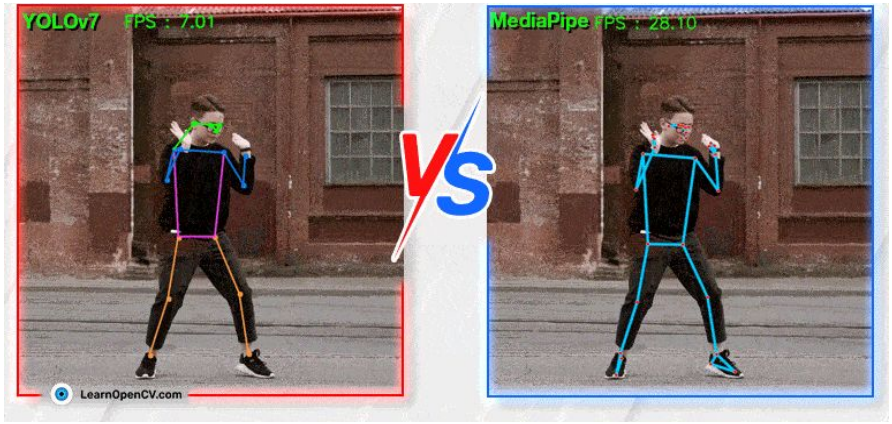
{EPITECH}

<SINCE 1999/>

< Robotique TEK5 · Day 3 · Visio · Part 4 />

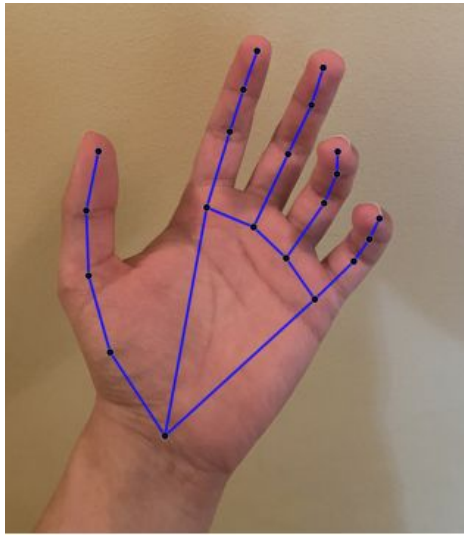
Humans: body, hands, face

< Body, hands, face />



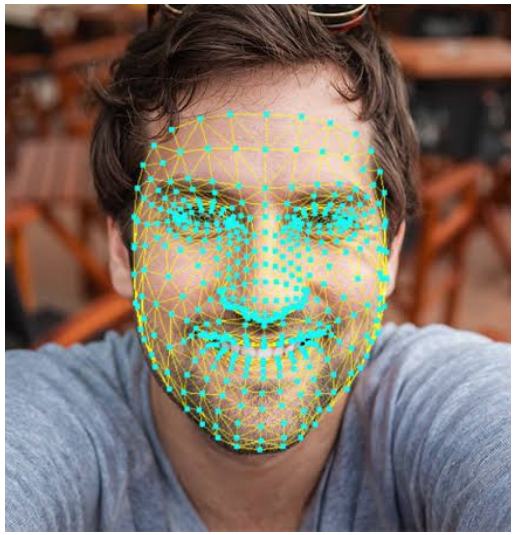
Body

`MediaPipe · RTMPose · ViTPose`



Hands

`MediaPipe Hands · HaMeR`

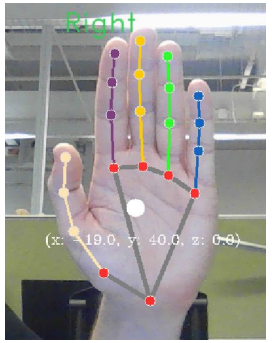


Face

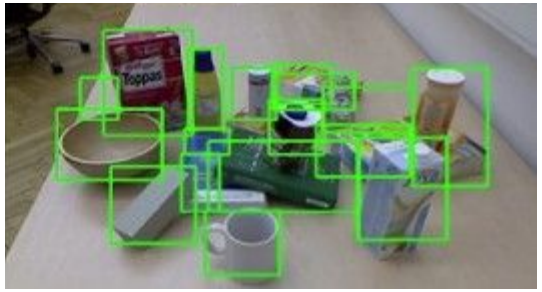
`MediaPipe Face Landmarker`

Start with MediaPipe: CPU, real time

< Ideas for phase 2 using perception />



Gestures
Hand teleop



Object selection



Safety stop / avoidance ?



Gaze



Learn from video



Voice - LLM - VLM



Failure detection